Finite fields and Fibonacci: Lecture 2

Counting and building finite fields

Bases turn algebra into counting

$$\mathbb{F}_7 \subset \mathbb{F}_{49}$$

Recall: scalars determine the combinations

In $\mathbb{F}_3^2$, let $u = (1, 2)$.

$$\operatorname{span}(u) = \{0u, 1u, 2u\} = \{(0, 0), (1, 2), (2, 1)\}.$$

With $e_1 = (1, 0)$ and $e_2 = (0, 1)$,

$$(a, b) = ae_1 + be_2.$$

How can we tell whether coordinates are unique?

A redundant spanning list

In $\mathbb{F}_3^2$, put $w = e_1 + e_2$.

The list e_1, e_2, w spans the space, but

$$w = 0e_1 + 0e_2 + 1w = 1e_1 + 1e_2 + 0w.$$

The difference gives a nontrivial relation:

$$e_1 + e_2 - w = 0.$$

Linear independence

Vectors $v_1, \ldots, v_d$ are linearly independent if

$$a_1 v_1 + \cdots + a_d v_d = 0 \quad \Longrightarrow \quad a_1 = \cdots = a_d = 0.$$

Then two expressions for the same vector have the same coefficients:

$$\Sigma\, a_i v_i = \Sigma\, b_i v_i \implies \Sigma(a_i - b_i) v_i = 0.$$

A basis gives existence and uniqueness

A basis is a linearly independent spanning list.

Every $v \in V$ has a unique expression

$$v = a_1 v_1 + \cdots + a_d v_d.$$

- Spanning gives existence of the coefficients.
- Independence gives uniqueness.

Build a basis one vector at a time

Start with no vectors; their span is $\{0\}$.

If the current span is not all of V, choose a vector outside it.

Adding that vector preserves independence.

In a finite space, this process must stop.

At stopping, the chosen vectors form a basis.

Could two bases have different lengths?

A basis gives unique coordinates.

But before defining dimension, we must prove that its length does not depend on the chosen basis.

Our key comparison:

$$\text{independent list length} \leq \text{spanning list length}.$$

Exchange one vector

Suppose $w_1, \ldots, w_m$ span V and $v \neq 0$.

$$v = a_1 w_1 + \cdots + a_m w_m.$$

Some coefficient is nonzero; relabel so $a_1 \neq 0$.

$$w_1 = a_1^{-1}\big(v - a_2 w_2 - \cdots - a_m w_m\big).$$

Thus $v, w_2, \ldots, w_m$ still span V.

An exchange in $\mathbb{F}_3^2$

Start with $e_1 = (1, 0)$ and $e_2 = (0, 1)$.

Insert $u = (1, 1) = e_1 + e_2$.

$$e_1 = u - e_2, \qquad u, e_2 \text{ still span.}$$

Now insert $v = (1, 2) = u + e_2$.

$$e_2 = v - u, \qquad u, v \text{ still span.}$$

Repeat the exchange

Let $v_1, \ldots, v_r$ be independent and $w_1, \ldots, w_m$ span V.

After inserting $v_1, \ldots, v_k$, the list still spans and has

$$v_1, \ldots, v_k, \quad m - k \text{ remaining old vectors.}$$

Express v_{k+1} in this list.

Some remaining old vector has a nonzero coefficient.

Otherwise $v_{k+1} \in \operatorname{span}(v_1, \ldots, v_k)$, contradicting independence.

An independent list cannot be longer

Every insertion replaces one remaining old vector.

After m insertions, the spanning list would be

$$v_1, \ldots, v_m.$$

If $r > m$, then v_{m+1} would be in their span, contradicting independence.

$$r \leq m.$$

This is the Steinitz exchange argument.

Dimension is well defined

Suppose B and C are bases, of lengths r and m.

- B independent, C spanning: $r \leq m$.
- C independent, B spanning: $m \leq r$.

Therefore $r = m$.

For a space with a finite basis, this common length is its dimension:

$$\dim_k V = r.$$

Count coordinate choices

Suppose $v_1, \ldots, v_d$ is a basis over $\mathbb{F}_p$.

$$v = a_1 v_1 + \cdots + a_d v_d, \qquad a_i \in \mathbb{F}_p.$$

Each of the d coordinates has p choices.

Uniqueness means that different choices give different vectors.

$$|V| = p^d.$$

Your turn: is this a basis?

In $\mathbb{F}_3^2$, let $u = (1, 1)$ and $v = (1, 2)$.

1. Solve $au + bv = (0, 0)$.
2. Express $(2, 0)$ as $au + bv$.
3. How many vectors are in $\mathrm{span}(u, v)$?

Write the two coordinate equations.

Basis exercise: answers

$$a + b = 0, \quad a + 2b = 0 \quad \Longrightarrow \quad b = 0, a = 0.$$

So u, v are independent.

$$(2, 0) = u + v.$$

Their span has $3^2 = 9$ elements, all of $\mathbb{F}_3^2$.

Thus u, v form a basis.

A field can be a vector space

Let K be a finite field of characteristic p.

It contains $\mathbb{F}_p$. Regard

- elements of K as vectors;
- elements of $\mathbb{F}_p$ as scalars;
- multiplication in K as scalar multiplication.

$$a(u+v) = au + av, \qquad a(bu) = (ab)u.$$

Every finite field has prime-power size

Finiteness gives a basis of K over its prime field $\mathbb{F}_p$.

$$|K| = p^d, \qquad p \text{ prime}, \quad d \geq 1.$$

- Characteristic gives the prime p.
- A basis over $\mathbb{F}_p$ gives the exponent d.

There is no field with 6 elements.

A field with 49 elements would have characteristic 7.

Build the missing square root

Over $\mathbb{F}_7$, 5 is not a square.

Introduce a symbol s with $s^2 = 5$, and use formal pairs

$$K = \{a + bs : a, b \in \mathbb{F}_7\}.$$

Two elements are equal exactly when their two coefficients agree.

$$(a + bs) + (c + es) = (a + c) + (b + e)s.$$

Multiply, then replace s^2 by 5

$$(a+bs)(c+es) = (ac+5be) + (ae+bc)s.$$

For example,

$$\begin{aligned}(2+s)(3+2s) &= 6+7s+2s^2 \\ &= 16+7s \\ &= 2.\end{aligned}$$

All coefficients are reduced modulo 7.

Why every nonzero element has an inverse

$$(a+bs)(a-bs) = a^2 - 5b^2.$$

For $(a,b) \neq (0,0)$, this denominator is nonzero:

- If $b = 0$, then $a^2 \neq 0$.
- If $b \neq 0$, its vanishing would give $(a/b)^2 = 5$.

Therefore

$$(a+bs)^{-1} = \frac{a-bs}{a^2-5b^2}.$$

A field with forty-nine elements

We have built a field K, which we call $\mathbb{F}_{49}$.

$\{1, s\}$ is a basis over $\mathbb{F}_7$.

Every element is a unique $a \cdot 1 + b \cdot s$.

$$|K| = 7^2 = 49, \qquad \dim_{\mathbb{F}_7} K = 2.$$

Its characteristic is 7.

$\mathbb{F}_{49}$ and $\mathbb{Z}/49\mathbb{Z}$ are different

	$\mathbb{F}_{49}$	$\mathbb{Z}/49\mathbb{Z}$
Elements	49	49
Characteristic	7	49
Every nonzero element a unit?	Yes	No

In $\mathbb{Z}/49\mathbb{Z}$, the nonzero residue 7 satisfies $7^2 = 0$.

In $\mathbb{F}_{49}$, seven copies of 1 already give 0.

Exit question

Suppose K is a field with 25 elements.

1. What is its characteristic?
2. What is its dimension over its prime field?
3. Why is K not $\mathbb{Z}/25\mathbb{Z}$?

Polynomial notation for the construction

$$\mathbb{F}_{49} = \mathbb{F}_7[X]/(X^2 - 5).$$

Two polynomials represent the same element when their difference is a multiple of $X^2 - 5$.

Polynomial division gives a unique remainder $a + bX$.

The class of X is our element s, and $s^2 = 5$.

What if the chosen number is already a square?

Try the same formal-pair construction over $\mathbb{F}_{11}$, with $s^2 = 5$.

Since $4^2 = 5$,

$$(s-4)(s+4) = s^2 - 16 = 0.$$

Both factors are nonzero formal pairs.

This construction gives a ring with zero divisors, not a field.

Your turn: calculate in the new arithmetic

In K, where $s^2 = 5$ and coefficients lie in $\mathbb{F}_7$, compute

$$(1 + 2s)(3 + s).$$

Then compute

$$(2 + s)(2 - s).$$

Distribute first; reduce afterward.

Two products

$$(1+2s)(3+s) = 3+7s+2s^2 = 13 = 6.$$

$$(2+s)(2-s) = 4-s^2 = 4-5 = 6.$$

Since $6^{-1} = 6$ in $\mathbb{F}_7$,

$$(2+s)^{-1} = 6(2-s) = 5+s.$$

The roots now exist

In $\mathbb{F}_{49}$, division by 2 is allowed, and

$$\alpha = \frac{1+s}{2}, \qquad \beta = \frac{1-s}{2}.$$

$$\alpha + \beta = 1, \qquad \alpha\beta = -1.$$

Both solve $X^2 - X - 1 = 0$.

Next: multiplicative orders, matrices, and the period.

A bridge to matrices

An invertible matrix has columns that form an ordered basis.

Counting invertible matrices = counting ordered bases.

We will count the column choices one at a time.

Gaussian elimination gives another proof

Suppose $w_1, \ldots, w_m$ span and $v_1, \ldots, v_r$ are independent.

Choose coefficients $v_i = \Sigma_{j=1}^{m} c_{ji} w_j$; let $C = (c_{ji})$.

If $r > m$, elimination in the homogeneous system $Cx = 0$ leaves a free variable, hence a nonzero solution.

$$\sum_{i=1}^{r} x_i v_i = \sum_{j=1}^{m} (Cx)_j w_j = 0,$$

contradicting independence.